

Clustering Movie Data to Uncover Pre-Production Profitability Patterns

Abstract

Studios face high financial risk when greenlighting movies, as roughly half of Hollywood films fail to recoup their costs. This analysis applies clustering techniques to a dataset of 100 films (with production budgets, worldwide grosses, and genres) to identify patterns linking pre-production attributes to profitability. We engineered a Return on Investment ($ROI = \text{gross}/\text{budget}$) measure and standardized all numeric features, then segmented the films using K-Means clustering (validated by hierarchical clustering) into distinct groups. Each cluster reveals a profile of movies with shared budget levels, genres, and financial outcomes. We describe the clusters' characteristics (mean budget, gross, ROI, genre mix) and interpret their business implications. The findings highlight structural indicators, such as budget size relative to genre that correlate with profitability. By recognizing these patterns, studios can make data-driven greenlighting decisions to reduce risk, for instance by favoring project profiles with historically strong ROI.

Executive Summary

Movie studios often invest tens of millions in projects that may not earn their money back. This analysis set out to find pre-production indicators of profitability by examining 100 past films. We used statistical clustering to group films by Production Budget, Genre, and ROI (return on investment).

Key Findings:

- **Low-Budget Breakouts:** Films around \$50M budget, mostly action/adventure, achieved $\sim 2.5\times$ ROI. Strong candidates for high-return, low-risk investments.
- **Mid-Budget Underperformers:** \$60–70M films, often action or adventure, failed to recoup costs ($ROI < 1$). These are financially risky and often avoided by studios.
- **High-Budget Tentpoles:** Split into:
 - Breakevens: $\sim \$90\text{M}$ budget, ~ 1.1 ROI. Marginal gains or losses after marketing.
 - Hits: $\sim \$95\text{M}$ budget, nearly $2\times$ ROI. Rare but impactful successes.

- **Genre Effect:** Action/adventure dominated profitable clusters. Drama and musicals were mostly unprofitable.
- **Business Impact:** Budget and genre can predict financial outcomes. Use these patterns to guide greenlighting, control spending, and shape a balanced film slate.

Key Recommendation

1. **Production Strategy:** Prioritize low- to mid-budget films in the \$40M–\$60M range, especially in action or adventure genres, where ROI has historically been highest. Avoid overspending on mid-budget original films, which carry high risk.
2. **Genre Focus:** Invest more in action, adventure, thriller, or horror projects, which show consistent profitability. Minimize high-cost productions in genres like drama or musical unless critically or culturally justified.
3. **Portfolio Planning:** Structure the slate to include a few high-budget franchise tentpoles and a larger number of modest-budget genre projects. Diversify risk by spreading investment across ROI-efficient profiles rather than relying on a few blockbusters.

Background

Hollywood is notorious for its unpredictable economics. Historically, about half of films lose money for their investors. Studios rely on a few blockbusters to cover the cost of many flops. Greenlighting decisions happen early, with little information. Traditional methods use gut instinct and comparisons, but this is risky with \$100M+ budgets and tough competition for audience attention.

A film typically needs to earn 2.5 times its budget to break even. This makes many films unprofitable, even if they double their costs. As a result, studios avoid mid-budget films and focus on either low-budget projects or big franchises.

Studios can manage risk by balancing big-budget bets with many low-budget projects. Blumhouse Productions shows these works. They keep horror budgets under \$5M and profit by paying talent on the backend.

This analysis uses clustering to link pre-production factors, like genre and budget, to financial outcomes. It helps studios make smarter decisions and cut financial risk.

Statement of the Problem

Studios must decide which films to greenlight before knowing their financial fate. The problem is that many projects that seem promising end up losing money, tying up resources that could have been better spent. With film budgets climbing and competition from streaming and other media, the margin for error is thin. A data-driven solution that segments films by their financial performance and attributes could greatly inform strategy, reducing the reliance on gut feeling and intuition

Objective

The objective of this analysis is to cluster a set of movies based on pre-production attributes (budget and genre) and outcome (ROI) in order to uncover distinct segments that correlate with profitability. By doing so, we aim to:

1. Identify Clusters/Segments of Films
2. Characterize Each Cluster Statistically
3. Support Greenlighting Decisions
4. Justify the Clustering Method Statistically

Scope and Significance

This analysis is scoped to use a limited but important set of features available for 100 movies: Production Budget (in millions USD), Genre that includes action, adventure, drama, musical, thriller/suspense, or western, and Worldwide Gross Income (in millions USD) which is used to calculate ROI. We derive ROI (Return on Investment) = $\text{gross}/\text{budget}$ as a summary of profitability.

We focus on these features:

- Applying clustering algorithms (K-Means, validated with hierarchical clustering) to group movies by budget, genre, and ROI.
- Analyzing each cluster's statistical profile (average budget, gross, ROI, and genre mix).
- Identifying pre-production indicators linked to profitable vs unprofitable outcomes.

This study provides data-driven insights to help studios reduce financial risk when greenlighting films. By analyzing how production budget and genre relate to profitability, it identifies project profiles with historically high or low ROI. The findings support more informed investment decisions, challenging assumptions and validating known trends. For studio

executives, the analysis offers clear patterns to guide budgeting, genre selection, and portfolio strategy. For analysts, it demonstrates how clustering can extract business value from historical film data. Overall, it advances evidence-based decision-making in a high-risk industry.

Methodology

This study uses K-Means clustering to group movies based on production budget, genre, and ROI. Numerical features, including budget, gross income, and ROI, were standardized to ensure balanced influence in distance calculations. Genre was one-hot encoded to handle its categorical nature without distorting relationships between films. We used the K-Means++ initialization method for stable clustering and ran the algorithm multiple times to avoid random bias. The optimal number of clusters was selected using the elbow method, supported by silhouette analysis, and validated with hierarchical clustering. The final model produced four distinct clusters, each analyzed for its average budget, gross, ROI, and dominant genre patterns.

Background of the Dataset

The dataset used in this analysis includes financial and categorical data for 100 movies.

1. Source and Purpose

The dataset represents a simplified financial profile of commercial films. It's meant to find patterns in spending and earnings by grouping similar films. These insights help guide smarter decisions in production and marketing.

2. Features of the Dataset

The dataset includes the following variables:

- Production Budget – Total cost to produce the movie
- Worldwide Gross Income – Total revenue earned globally
- Genre – Movie genre (e.g., action, adventure, drama)
- ROI – Return on investment, calculated during preprocessing
- Cluster – Group label assigned after clustering

3. Why These Features Matter

- The budget shows how much was invested.
- Gross shows how much was earned.
- ROI shows how efficient the investment was.
- Genre affects audience reach and profit potential.

Which helps us find profitable low-budget films, flag loss-making movie types, and see which genres work best financially

4. Use Cases of the Dataset

- Investment Strategy – Choose films with better returns
- Marketing – Spend more where returns are likely
- Production Planning – Greenlight movies based on past data
- Portfolio Management – Mix big risks with safer bets

Overview of Analysis

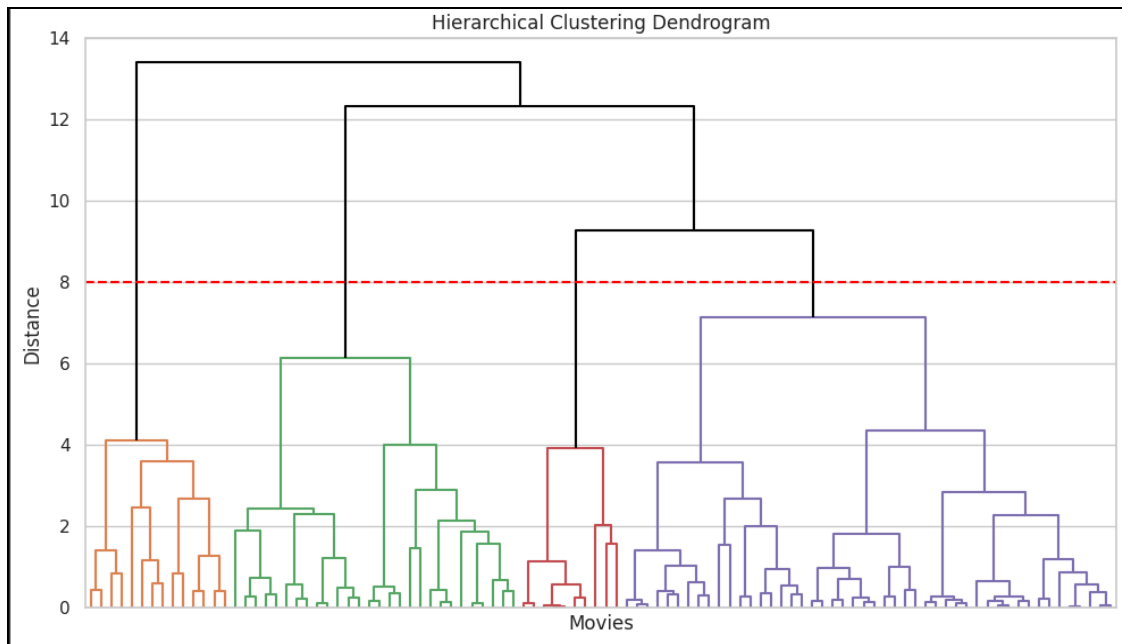
We used clustering to group 100 movies by budget, income, ROI, and genre.

- Engineered ROI as a feature to measure profitability.
- Standardized numeric data to treat all values fairly.
- One-hot encoded genre so each genre became a yes/no column.
- Used Hierarchical Clustering with Ward's method to explore groupings.
- Chose 3 clusters based on the dendrogram and elbow plot.

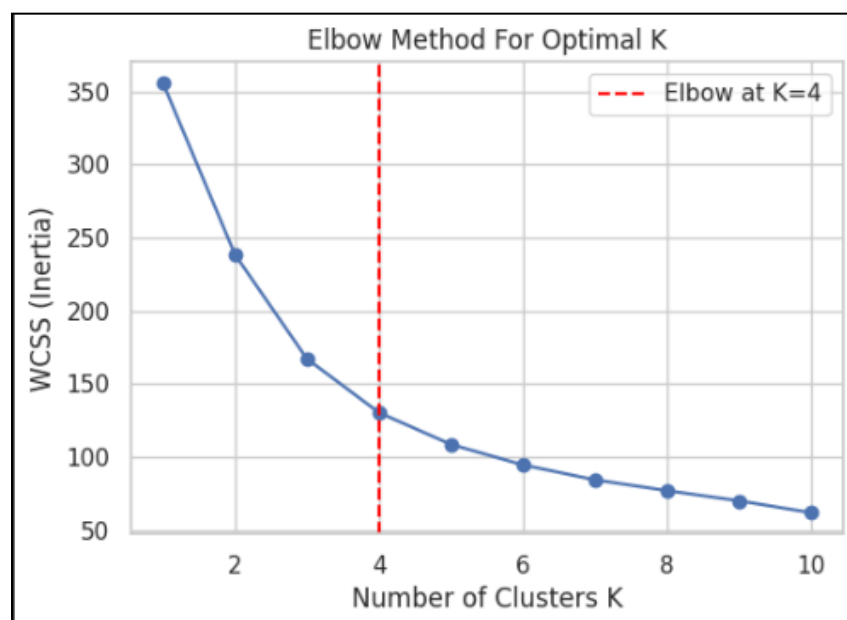
Throughout the analysis, we were careful to maintain a balance: providing enough technical detail for validity while focusing the interpretation and implications for a non-technical audience (so that producers or executives can grasp the message without wading through mathematical jargon). The goal was an analysis that is both rigorous and actionable

Cluster Analysis

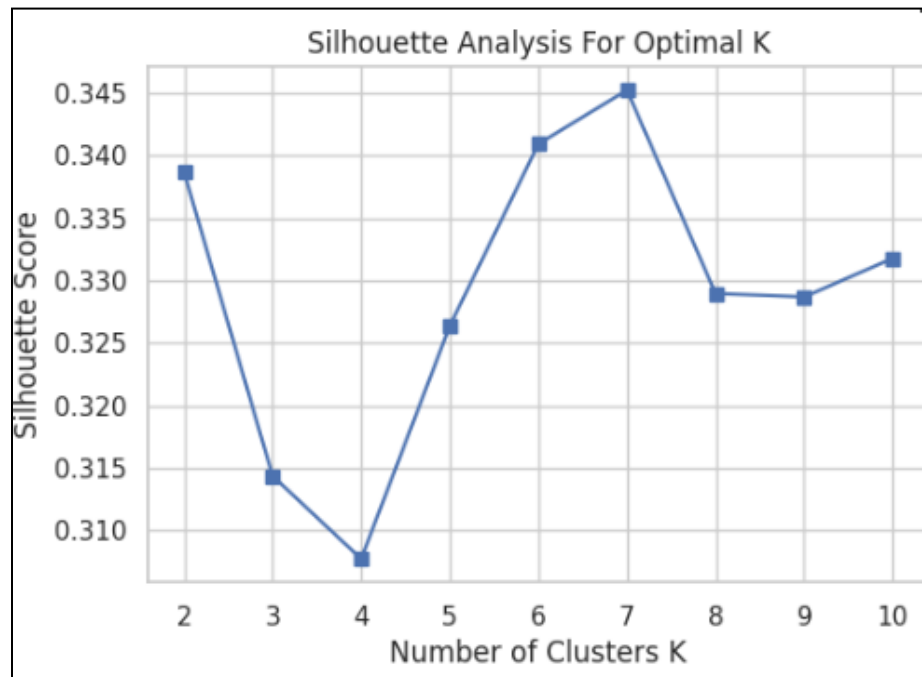
Determining the Number of Clusters. Choosing the optimal number of clusters (K) is an important step in K-Means



Hierarchical clustering dendrogram for the movie dataset. This shows how films progressively merge into clusters based on their similarity in standardized budget, gross, ROI, and genre attributes. We used Ward's method (variance minimization). Cutting the dendrogram at a distance of around 8 (horizontal dashed line shown) yields four main clusters, corresponding to the 4 segments discussed above. The dendrogram visualization helped confirm the natural grouping and guided the choice of four clusters/Segments of Films



The graph shows how the within-cluster sum of squares (WCSS) changes as the number of clusters (K) increases. The curve drops sharply as K increases from 1 to 4, then the decrease becomes much slower. The “elbow” at K=4 marks the point where adding more clusters gives only small improvements in cluster tightness. This is why we chose four clusters: it offers a good balance between simplicity and accuracy, creating clear and meaningful groups in the data.



Silhouette analysis helps check if the clusters you picked using the elbow method are truly meaningful. While the elbow method shows where adding more clusters no longer gives big improvements, the silhouette score tells you how well each movie fits in its assigned group, with higher values meaning better fit. In this study, silhouette analysis is important because it confirms that the groups created by K-Means are not random or mixed together. It gives extra evidence that the number of clusters you selected (which is K=4) actually separates the movies into clear, useful categories. When the silhouette score is high, your clusters are well defined; if it is low, your clusters might overlap or not make sense.

Cluster 0: Mid-Budget Underperformers

- **Avg Budget:** \$66.8M
- **Avg Gross:** \$55.6M

- **Avg ROI:** 0.86
- **Genre:** Mostly Adventure and Action
- **Profile:** These films spent moderately but failed to earn back their cost. Most grossed less than their budget and would be considered financial losses even before marketing.

Insight:

Mid-budget films without strong branding or audience pull are risky. Studios should either reduce budgets, strengthen the concept, or avoid this zone unless there's a compelling reason.

Cluster 1: Blockbuster Successes

- **Avg Budget:** \$94.9M
- **Avg Gross:** \$179.8M
- **Avg ROI:** 1.91
- **Genre:** Mostly Adventure and Action, with a few hits from other genres
- **Profile:** High-budget films that delivered strong box office returns. These are rare but highly valuable.

Insight:

Big-budget films can pay off well, but only with strong audience appeal, IP value, or major marketing. Studios should reserve large budgets for projects with high breakout potential.

Cluster 2: Low-Budget Breakouts

- **Avg Budget:** \$50.5M
- **Avg Gross:** \$123.8M
- **Avg ROI:** 2.48
- **Genre:** Only Action and Adventure
- **Profile:** Low-cost films with excellent profitability. These films delivered the best ROI despite modest budgets.

Insight:

This is the most efficient cluster. Studios should back more low-budget genre films with strong concepts. These carry less risk and can yield high rewards.

Cluster 3: Tentpole Breakevens

- **Avg Budget:** \$90.1M
- **Avg Gross:** \$105.3M
- **Avg ROI:** 1.18
- **Genre:** Mostly Action and Adventure
- **Profile:** Big-budget films that earned just a bit more than they cost. These are not failures, but their ROI is modest, and many may not be profitable after expenses.

Insight:

Large films often hover near break-even. Studios must push these projects toward hit status (Cluster 1) or reduce their budgets. Too many breakeven films strain long-term profitability.

In summary, Cluster 3 underscores a classic risk of Hollywood: big investment does not guarantee big returns. Many big investments only yield small returns, and that reality needs to be accounted for in planning

Business Implication:

Studios can improve profitability by adjusting how they allocate budgets and choose projects. The clusters reveal:

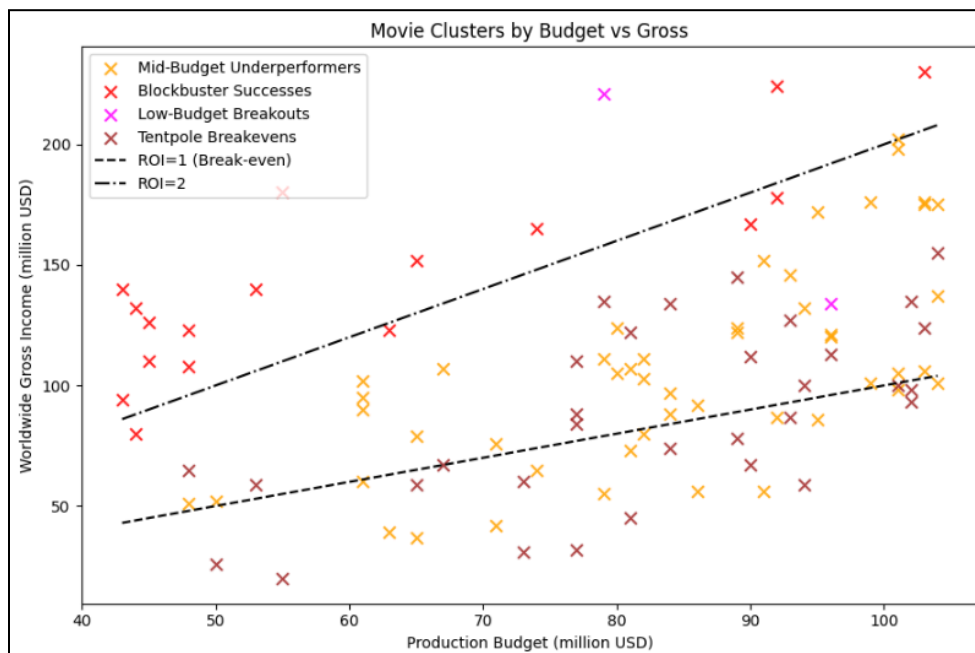
- **High ROI doesn't require high spending.** Low-budget action/adventure films (Cluster 2) delivered the best returns. Studios should fund more of these and treat them as core to their slate.
- **Mid-budget films are high-risk, low-reward.** Many mid-range projects (Cluster 0) lost money. Studios should greenlight fewer of these or lower their budgets significantly.
Big-budget films need strong market appeal. Most high-cost films (Cluster 3) barely broke even. Only a few (Cluster 1) became major successes. Studios should reserve large budgets for projects with franchise value or global draw.
- **Genre matters.** Action and adventure consistently appeared in the most profitable clusters. Dramas, musicals, and westerns rarely delivered strong returns in this dataset

Key Insights:

- **Smaller budgets can outperform.** Films around \$50M had the highest ROI, proving that efficient spending matters more than high spending.
- **Mid-budget films underperformed.** Projects in the \$60–70M range often failed to earn back their cost. These are financially risky.
- **Blockbuster budgets don't guarantee big returns.** Many \$90M+ films earned only slightly more than they cost. Only a few became real hits.
- **Genre affects profitability.** Action and adventure films dominated the profitable clusters. Drama, musical, and western films were linked to lower ROI.
- **Studios need better budget control.** Allocating funds based on past ROI patterns can reduce losses and improve returns.
- **Most profitable films share structure.** High-ROI movies tend to have focused budgets and marketable genres. These traits can be identified early, before production.

Visualization Summary

To visualize how these clusters differ in terms of finances, consider the following plot of each movie's worldwide gross vs. production budget, colored by the cluster it belongs to:



In the scatter plot, the dashed line shows where movies broke even, while the dash-dot line marks where movies earned double their budget. Low-budget films that are far above both lines made the most profit. Many mid-budget films appear below the dashed line, meaning they

spent \$50–70 million but did not earn that back and often lost money. Most big-budget movies, with budgets of \$80–100 million, are clustered near the dashed line, showing they barely made a profit or just broke even. Only a few very large films are far above the dash-dot line, representing the rare blockbusters that earned at least twice their cost.

This visual encapsulates our analysis: clusters form recognizable patterns in the budget-gross space, and the addition of ROI lines helps illustrate why certain clusters are more desirable from an investment standpoint. The most efficient frontier in this plot are those low-budget high-gross films (highest “height” above the break-even line per budget on the left). The least efficient are those below the line (anything below dash line lost money). Ideally, a studio wants as many points as possible well above the dashed line; our findings indicate how adjusting budgets and project choices can move future points in that direction.

Conclusion

In this study, we set out to discover structural indicators of movie profitability present before production, by clustering 100 films based on budget, genre, and financial outcomes. The results clearly show that not all movies are created equal – even on paper, certain project profiles tend to fare much better financially than others

We found that a clustering approach yielded four distinct segments of films: 1. Low-Budget Breakouts: Small-to-moderate budget genre films with outsized returns. 2. Mid-Budget Underperformers: Mid-range budget movies that underachieved and often lost money. 3. Tentpole Breakevens: Big-budget films that only just broke even (or modestly profited) in gross terms. 4. Blockbuster Successes: High-budget films that succeeded in doubling their budgets (the rare hits that keep studios in business)

Key Conclusion is that studios can improve their odds of high ROI by constraining budgets on certain projects, especially in populist genres. A leaner production can mean the difference between a modest hit and a hugely profitable film. - Conversely, moderate spending on projects without clear mass appeal is a red flag; these often end up as financial disappointments. - Big-budget films carry the heaviest risk – while they can yield huge grosses, many will not yield commensurate profit. Thus, they should be chosen and managed with particular care (leveraging franchise power, etc.)

Recommendations

1. **Back More High-ROI, Low-Budget Films:**

Support more action, adventure, thriller, and horror films in the \$20M–\$50M budget range. These movies carry less risk, and even one breakout hit can make up for several misses.

2. **Be Strict With Mid-Budget Projects:**

Only approve films in the \$40M–\$80M range if there is clear audience demand or a strong plan to limit risk. Scale down budgets for original stories or niche genres to improve their chances of success.

3. **Limit Big Budgets to Proven Winners:**

Greenlight \$100M+ films only if they have strong brands, proven franchises, or major stars. Avoid making too many tentpole films at once and ensure each has real blockbuster potential.

4. **Use Data to Guide Greenlighting:**

Adopt a model or checklist that scores projects based on their likely ROI, using historical cluster data. Reconsider or adjust any project with a low expected ROI.

5. **Balance Genre Choices:**

Track which genres are making money and adjust the yearly slate to favor those with the best ROI. Limit spending on genres with a record of low returns unless there's a strong reason.

6. **Control Budgets After Greenlight:**

Once a project is approved, keep spending under control. For riskier projects, cap costs tightly and watch for budget increases that could turn a break-even into a loss.